

AI-POWERED INTELLIGENT SALES FORECASTING PLATFORM USING PREDICTIVE ANALYTICS GROUP-1

Team Members & Roles:

No.	Team Member	Role
1	Ritika Kushwaha	Project Lead / Scrum Master, Backend Developer
2	Aditi Bhosure	Database / DevOps Engineer
3	Samraiyam Nallagorla	AI/ML Engineer
4	Tejashwini M	Frontend Developer
5	Rushikesh Arepalli	QA / Test Engineer

Project Overview:

SalesGenie AI is an AI-based sales assistant designed to help sales teams manage leads and better understand potential customers. It brings lead management, lead analysis, lead scoring, AI-powered outreach, recommendations, CRM integration, and sales analytics together in one platform.

The project is intended to reduce the manual effort involved in handling leads. By analyzing the available information, the platform can provide useful insights and recommend an appropriate next action. The application is being developed module by module so that each feature contributes to the overall sales workflow.

Project Objective:

The main objective of SalesGenie AI is to create an intelligent sales assistant that helps sales teams manage and work with leads in a simpler, faster, and more organized way. The platform is designed to help teams:

- Manage and organize leads in one place.
- Store and access prospect information easily.
- Analyze lead and company details using AI.
- Generate lead qualification scores to understand lead potential.
- Create personalized outreach messages based on lead information.
- Suggest suitable follow-up actions for sales teams.
- Provide useful sales insights through a centralized platform.

Project Modules:

The SalesGenie AI project is divided into six main modules. Each module focuses on a different part of the sales process, and together they form the complete platform.

Module 1 - Lead Management & Prospect Database

Manages leads and prospects in one place, including adding, viewing, updating, searching, deleting, and importing lead information.

Module 2 - Lead Intelligence & Company Analysis

Uses AI to understand available lead and company information and provide useful insights about a lead.

Module 3 - AI Outreach Generation

Creates AI-generated emails and outreach messages from the lead information and industry so that communication is relevant and personalized.

Module 4 - Lead Scoring & Recommendation Engine

Evaluates leads, provides a lead score, and recommends follow-up timing, communication channel, and content strategy.

Module 5 - Conversation Intelligence & CRM Integration

Planned to connect SalesGenie AI with CRM systems and support conversation summaries and data synchronization.

Module 6 - Dashboard & Sales Analytics

Provides a central dashboard for sales information and insights such as pipeline, conversion-related metrics, and follow-up activities.

Milestone 2 Delivery Summary:

Milestone	Duration	Development Period	QA / Review	Focus Modules	Acceptance Focus	Status
Milestone 2	Day 15-20	04-Aug-2026 to 10-Aug-2026	11-Aug-2026	Module 3: AI Outreach Module 4: Lead Scoring & Recommendation Engine	Lead scoring target $\geq 85\%$; personalized outreach across multiple industries	Signed Off

Milestone - 2: QA Report

Testing Overview:

Milestone 2 was the second major testing checkpoint of the SalesGenie AI project. The main focus at this stage was to verify the AI Outreach Generation and Lead Scoring & Recommendation Engine features before moving to the next development phase.

The milestone covered the AI email and outreach generation service, the Streamlit Outreach tab with edit-before-send support, the lead scoring and conversion probability model, and the recommendation engine. It also included the Lead Score and Outreach Strategy cards integrated into the Outreach interface.

QA activities included functional testing, lead scoring validation, industry outreach testing, integration testing, and end-to-end testing. The completed workflow was reviewed on Day 20, and Milestone 2 was completed and signed off.

Testing Objectives:

The main goal of Milestone 2 testing was to make sure that the new AI-assisted outreach and lead-prioritization features were working correctly and could be used together in one workflow. The testing focused on the following areas:

1. To verify that AI outreach content could be generated using the available lead information and industry prompt templates.
2. To check that the Outreach tab displayed the generated content and allowed the user to review or edit it before sending.
3. To validate that the lead scoring and conversion probability functionality produced a score for the supplied lead information.

4. To verify that the recommendation engine returned useful guidance for follow-up timing, channel mix, and content strategy.
5. To check that Lead Score and Outreach Strategy cards were correctly integrated into the Outreach UI.
6. To test outreach generation across three or more industries.
7. To validate the complete frontend, backend, AI, and recommendation workflow before milestone sign-off.

Testing Scope:

The testing scope for Milestone 2 was mainly divided into the following areas.

AI Outreach Generation

- Generating AI email/outreach content from lead details.
- Checking industry-based prompt template handling.
- Reviewing whether the generated content was shown in the Outreach interface.
- Checking the edit-before-send flow.

Lead Scoring

- Generating a lead score and conversion probability from supplied lead information.
- Reviewing the lead-scoring target of $\geq 85\%$ as part of the milestone validation.
- Checking that the score was available to the Outreach workflow.

Recommendation Engine

- Generating recommendations for follow-up timing.
- Reviewing suggested communication channel mix.
- Reviewing recommended content strategy.

Integration

- Checking Lead Score and Outreach Strategy cards in the Outreach UI.
- Checking the end-to-end workflow from lead information through AI output, score, recommendation, and UI display.

Testing Environment & Tools:

The following tools and technologies were used or established for the project testing activities.

Area	Tool / Technology
API Testing	Postman
Backend	FastAPI
Database	PostgreSQL
ORM	SQLAlchemy
Frontend	Streamlit
AI Integration	Gemini / AI API
Automated Testing	PyTest
Version Control	Git
Manual Testing	Application UI / Browser

Testing Approach:

Milestone 2 used a combination of manual, functional, integration, and end-to-end testing. The AI-generated outreach, lead score, and recommendations were checked through the application workflow. Industry-based examples were used to review the outreach feature, while the Lead Score and Outreach Strategy cards were verified within the Streamlit interface.

Test Cases:

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-11	AI Outreach	Generate outreach using lead details and an industry prompt template	Personalized outreach content should be generated	Generated content was available for review	Pass
TC-12	AI Outreach	Review and edit generated outreach before send	User should be able to review/edit content in Outreach tab	Edit-before-send flow was validated	Pass
TC-13	AI Outreach	Generate outreach for multiple industries	Content should adapt to industry prompt templates	Industry outreach testing was completed for 3+ industries	Pass
TC-14	Lead Scoring	Generate lead score and conversion probability	Lead score and conversion probability should be produced	Scoring functionality was validated	Pass
TC-15	Lead Scoring	Validate milestone lead scoring target	Scoring target should be reviewed against $\geq 85\%$ acceptance target	Validated during Day 20 QA review	Pass
TC-16	Recommendation Engine	Generate follow-up recommendations	Timing, channel mix, and content strategy should be returned	Recommendations were available in the workflow	Pass
TC-17	Integration	Display Lead Score card in Outreach UI	Lead Score card should be displayed correctly	Card integration was validated	Pass
TC-18	Integration	Display Outreach Strategy card in Outreach UI	Outreach Strategy card should be displayed correctly	Card integration was validated	Pass
TC-19	End-to-End	Run the complete outreach, score, and recommendation flow	Components should communicate and show the required results	End-to-end workflow was validated	Pass

Test Execution Metrics:

The following table provides a concise QA execution summary for the documented Milestone 2 test scenarios. Functional, industry outreach, integration, and end-to-end validation were completed before the milestone sign-off.

Metric	Result
Total Test Cases	9
Test Cases Executed	9
Passed - Initial Execution	9
Failed - Initial Execution	0
Blocked	0
Initial Pass Percentage	100%

Metric	Result
Defects Identified	0
Defects Resolved	Not Applicable
Defects Retested	Not Applicable
Final Passed After Retesting	9
Final Pass Percentage	100%

Defect Summary:

No open or blocking defects remained at the time of the Milestone 2 QA review. The documented test scenarios completed successfully, allowing the milestone to proceed to sign-off.

Defect ID	Module	Issue	Severity	Status
N/A	Module 3 / Module 4	No open defects recorded at milestone sign-off	Not Applicable	Closed

Severity

- Critical: Major functionality is completely blocked.
- High: Important functionality is seriously affected.
- Medium: A specific feature is affected but the application can still be used.
- Low: Minor UI or usability issue.

Defect Resolution & Retesting:

The final QA review did not identify any open issues requiring defect retesting. The completed functional, integration, and end-to-end checks supported the final sign-off decision.

Defect ID	Issue	Retest Result	Final Status
N/A	No open defects requiring retest	Not Applicable	QA Complete

Summary

The documented Milestone 2 scenarios completed successfully. No open defects remained during the final QA review, and the milestone was approved for sign-off.

Integration Testing:

Integration testing was completed to make sure that the AI Outreach, Lead Scoring, Recommendation Engine, and Outreach UI were working together properly. The complete flow checked was:

Frontend -> FastAPI Backend -> AI / Scoring / Recommendation Services -> Backend -> Frontend

Areas Checked

- Whether the Outreach interface passed the required lead information to the backend.
- Whether the backend triggered the AI outreach, lead scoring, and recommendation workflow correctly.
- Whether generated outreach content, scores, and recommendations were returned to the application.
- Whether Lead Score and Outreach Strategy cards displayed the returned information correctly.
- Whether the complete workflow could be validated end to end.

AI Outreach / Lead Scoring / Recommendation Validation:

Milestone 2 validation focused on whether the three AI-assisted features worked together and produced results that could support the outreach workflow.

AI Outreach Validation

- The system was checked for generation of personalized outreach content using industry prompt templates.
- Outreach generation was tested across three or more industries.
- Generated content was reviewed in the Streamlit Outreach tab, including the edit-before-send step.

Lead Scoring Validation

- The lead scoring and conversion probability model was included in functional validation.
- The Day 20 QA review checked the milestone lead-scoring acceptance target of $\geq 85\%$.
- No actual measured accuracy value was supplied; this report does not state one.

Recommendation Engine Validation

- The recommendation engine was checked for follow-up timing, channel mix, and content strategy guidance.
- The returned strategy was reviewed as part of the integrated Outreach workflow.

QA Observation

The validation focused on both availability and workflow relevance: whether the application could generate outreach, score leads, present recommendations, and display the related cards in the Outreach UI. Functional, industry outreach, integration, and end-to-end testing were all included in the Day 20 review.

Overall QA Results:

Testing Area	Result
AI Outreach Generation	Passed
Outreach UI and Edit-Before-Send Flow	Passed
Industry Outreach Testing (3+ Industries)	Passed
Lead Scoring Validation	Passed
Recommendation Engine Validation	Passed
Lead Score and Outreach Strategy Card Integration	Passed
Integration Testing	Passed
End-to-End Testing	Passed
Milestone 2 QA Review / Sign-off	Completed and Signed Off

Overall QA Summary

Milestone 2 testing covered AI Outreach Generation, Lead Scoring, the Recommendation Engine, the Streamlit Outreach interface, and their complete integrated workflow. The QA review included functional testing, lead-scoring validation, industry outreach testing, integration testing, and end-to-end testing.

The outreach feature was reviewed across three or more industries, and the milestone lead-scoring acceptance target of $\geq 85\%$ was included in the Day 20 validation. The documented test scenarios completed successfully, with no open defects at the time of final review.

Based on the completed validation and Day 20 review, Milestone 2 was approved and signed off.

Conclusion:

Milestone 2 QA was completed to verify that the AI Outreach Generation and Lead Scoring & Recommendation Engine features were working properly and communicating correctly with the Outreach interface. The QA activities covered functional testing, industry-based outreach testing, lead-scoring validation, recommendation validation, integration testing, and end-to-end testing.

The completed review confirmed that the AI-generated outreach flow, lead score, recommendation strategy, and Outreach UI cards had been validated for the milestone. Milestone 2 was completed and signed off, providing a verified base for the next phase of SalesGenie AI development.

Milestone Sign-off:

Milestone	QA / Review Date	Validation Coverage	Final Status
Milestone 2 - AI Outreach Generation and Lead Scoring & Recommendation Engine	11-Aug-2026	Functional testing, lead scoring validation, industry outreach testing, integration testing, and end-to-end testing	Completed and Signed Off